University Enterprise Edition | ASP.NET Core (.NET 10.0) | Google Gemini AI Integrated | UGC 4-Quadrant Framework | English & Urdu Bilingual

EduDigiClass LMS — Comprehensive Enterprise User Manual

The definitive operational standard and end-to-end user guide for University Administrators, Faculty Instructors, and Student Learners. Covers curriculum hierarchy, Zoom live lectures, AI Question Paper generation, Gemini automated evaluation, UGC 10-point grading, and login-based attendance telemetry.

EXECUTIVE SUMMARY

Institutional E-Learning Framework & Architecture

EduDigiClass is an enterprise-grade digital learning platform designed to fulfill university-level statutory requirements under the University Grants Commission (UGC) and Distance Education Bureau (DEB) guidelines. The platform integrates academic curriculum delivery, virtual classroom teleconferencing, artificial intelligence evaluation engines, and automated attendance logging into a unified multi-tenant ecosystem.

🏛️

Super Admin / Management

Total governance over degree programs, course catalogs, faculty mappings, semester windows, and institutional grading policy configurations.

👩‍🏫

Faculty & Evaluators

Curriculum authoring, Zoom live scheduling, AI-assisted question paper generation, rubric-guided descriptive assessment evaluation, and performance monitoring.

🎓

Student Learners

Access to UGC 4-Quadrant course resources, interactive class calendars, timed online examinations, real-time attendance logs, and printable grade cards.

💡 **Single Source of Truth (SSOT) Architecture**

All academic course structures (Degree Programs, Semesters, Modular Subjects, and Credit Weightages) authored in EduDigiClass serve as the authoritative baseline across the university ecosystem, directly synchronizing with the central ERP Examination Ledger and CRM Admissions repository.

ACCESS GOVERNANCE

System URLs, Ports & Default Credentials

The platform operates on Port **82** under IIS / Kestrel reverse proxy on the production VM host (`20.193.144.80`). Below are the dedicated endpoint routes and seeded university accounts:

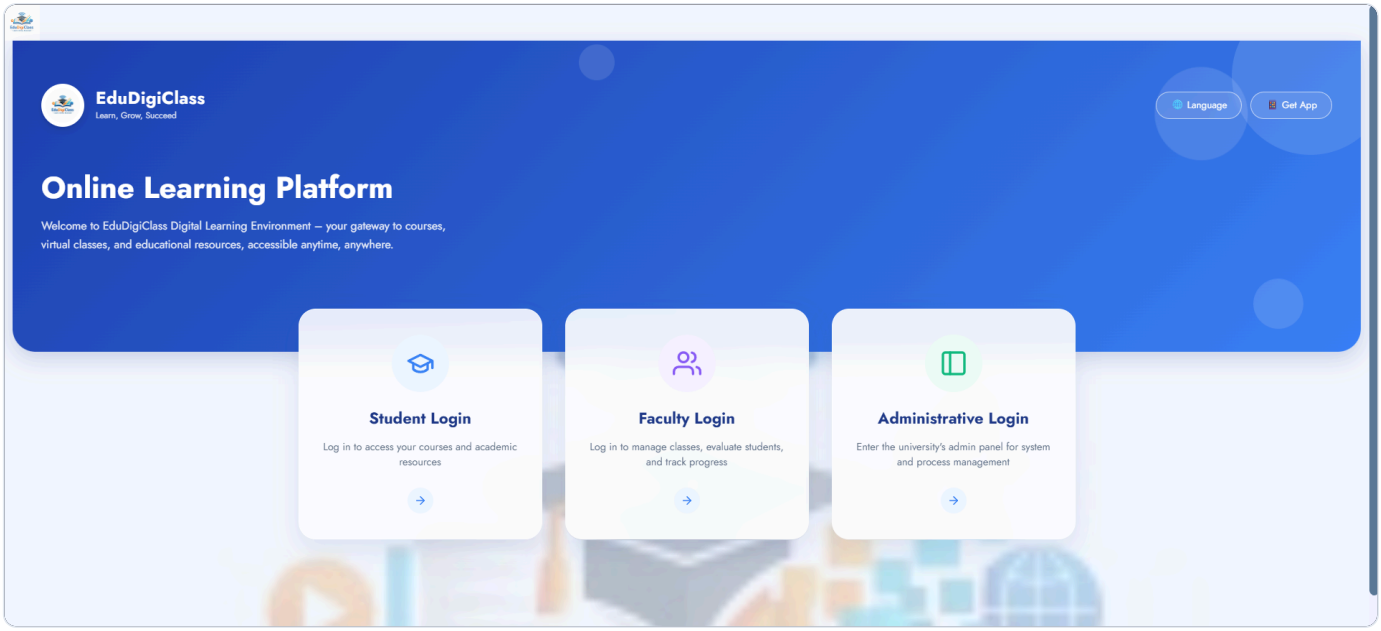
ROLE	PRIMARY PORTAL URL	DEFAULT LOGIN ID	DEFAULT PASSWORD	ACCESS PRIVILEGE LEVEL
Super Admin	`http://20.193.144.80:82/Admin/User`	`admin@TMS.com`	`Admin@123`	Full Institutional Root Access, User Deletion, Policy Control
Faculty / Teacher	`http://20.193.144.80:82/InternalAssessment/FacultyDashboard`	`faculty@edudigi.com`	`Faculty@123`	Course Content Uploads, Question Papers, Grading, Live Zoom
Student Learner	`http://20.193.144.80:82/MyCourses`	`student@edudigi.com`	`Student@123`	Learning Hub, Video Player, Quiz Attempts, Attendance Ledger

CHAPTER 1 • AUTHENTICATION GATEWAY

Multi-Role Landing Page & Secure Entry

EduDigiClass provides a unified, branded entry portal where users select their designated institutional role to enter their tailored workspace.

Figure 1: Online Learning Platform Gateway & Role Selector `Screenshot 2026-10-19 112709.png`



Unified authentication hub displaying direct entry cards for Student Login, Faculty Login, Administrative Login, Language selector, and Mobile App access.

Control Elements & Interface Breakdown

INTERFACE COMPONENT	ELEMENT TYPE	FUNCTIONAL PURPOSE & BEHAVIOR
Student Login Card	Primary Action Card	Routes directly to the student portal authentication desk (`/Auth/Login`). Grants access to course modules, study materials, live calendar, and assessments.
Faculty Login Card	Primary Action Card	Authenticates verified instructional staff to the Faculty Command Workspace (`/InternalAssessment/FacultyDashboard`) for classroom delivery and grading.
Administrative Login Card	Primary Action Card	Secure gateway into the university's institutional control console (`/Admin/User`) for governance, user mapping, and compliance audits.
Language Selector	Interactive Header Toggle	Instant bilingual switching between 🇬🇧 English and 🇮🇳 اردو with dynamic Right-to-Left (RTL) styling across all portals.
Get App Button	Header Utility	Provides direct QR codes and download links for the native Android and iOS mobile e-learning applications.

Bilingual (English / Urdu) RTL Support

EduDigiClass is equipped with full bilingual localization:

1. **Language Toggle Click**

   Click the language pill in the top header. Select **اردو (Urdu)** or **English**.

2. **Automatic RTL Layout Transformation**

   The entire interface mirrors horizontally. The sidebar shifts to the right, text aligns Right-to-Left, and typography switches dynamically without page reloads.

3. **Session & LocalStorage Persistence**

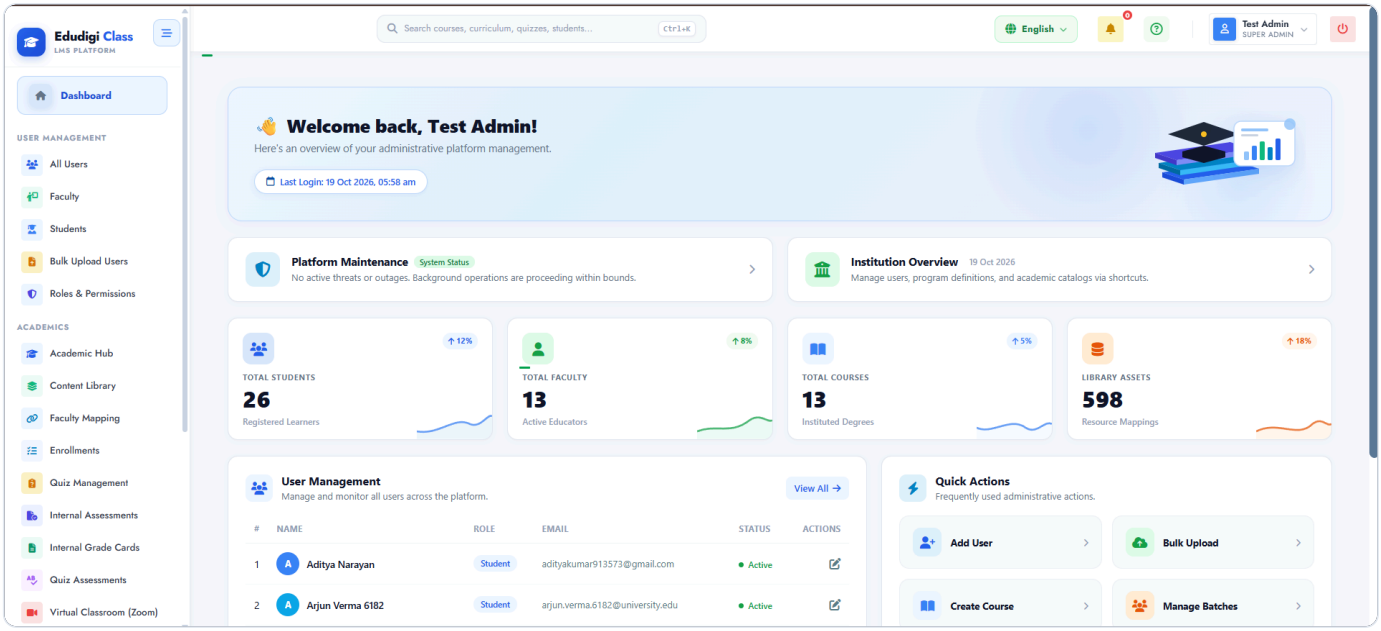
   The user's preferred language choice is stored in both HTTP cookies and browser LocalStorage, persisting across device reboots and navigation.

CHAPTER 2 • SUPER ADMIN OPERATIONS

2.1 Institutional Command Dashboard

The Super Admin Dashboard provides central oversight of institutional health, active learner enrollments, faculty counts, and academic catalogs.

Figure 2: Super Admin Platform Management Dashboard `Screenshot 2026-10-19 112930.png`



Executive dashboard displaying real-time counters, user metrics, quick navigation shortcuts, and administrative action centers.

Universal Global Search (Ctrl + K)

Instant search overlay accessible via `Ctrl + K`. Search across courses, syllabi, quizzes, faculty members, and students with sub-millisecond autocomplete.

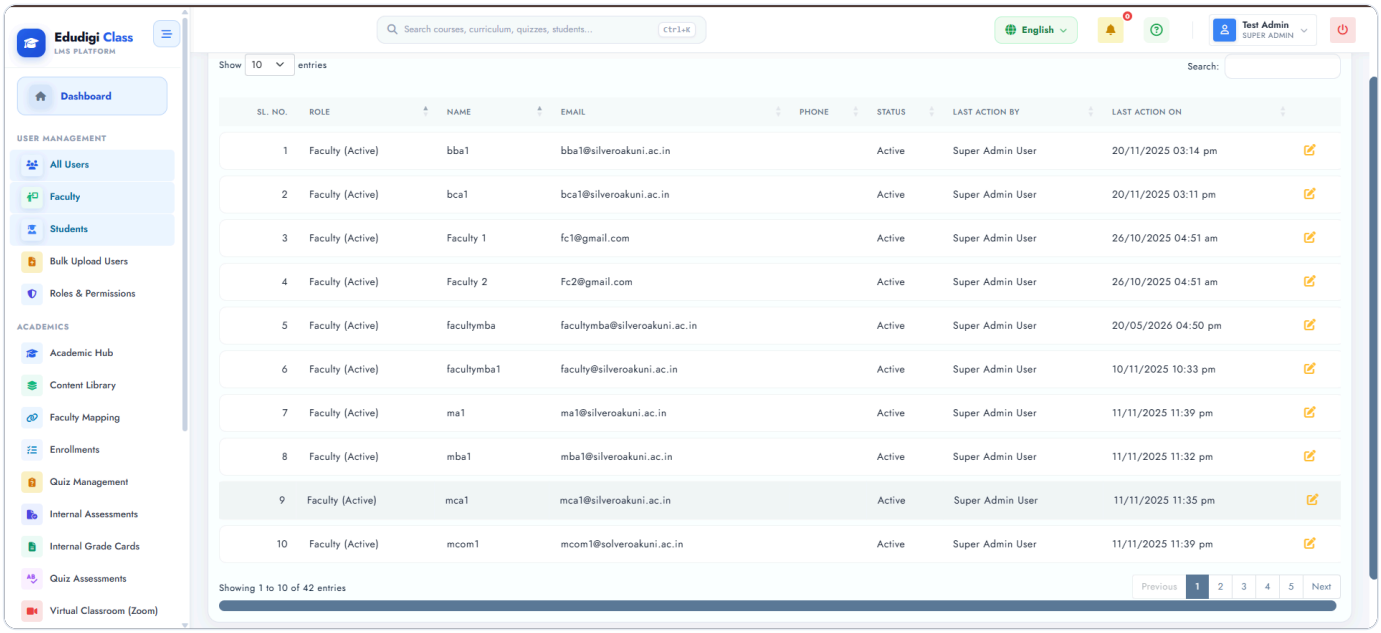
Role-Based Access Control (RBAC)

Granular permissions matrix for Super Admin, Academic Dean, Exam Coordinator, Faculty, and Student roles.

2.2 User Directory & Account Governance

Administrators manage user lifecycles, assign institutional roles, reset credentials, and activate/deactivate accounts.

📸 Figure 3: User Management Directory Table `Screenshot 2026-10-19 113026.png`



Data table showcasing user accounts, pagination controls, role classification badges (Faculty Active, Student, Admin), search filters, and action triggers.

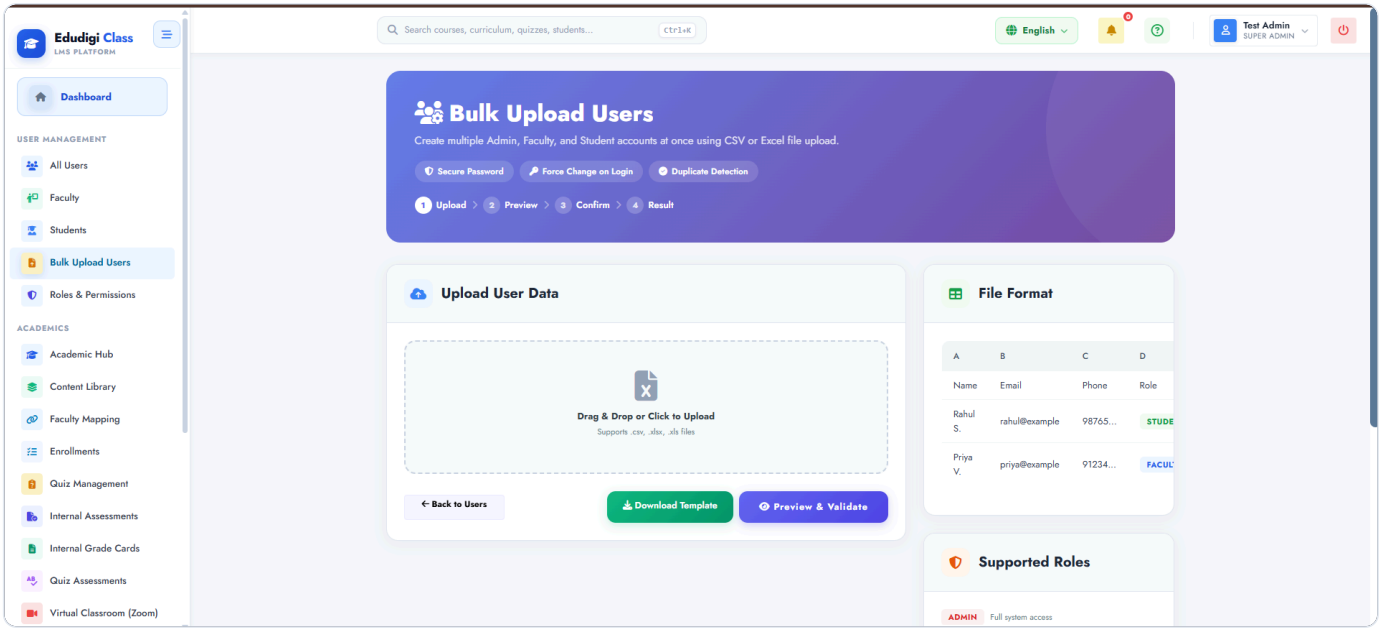
Operational Controls & Action Buttons

TABLE COLUMN / BUTTON	FUNCTION	ADMINISTRATOR GUIDANCE
SL. NO.	Sequential Record Index	Provides paginated ordering across total system user accounts (e.g. 1 to 10 of 42 entries).
ROLE	Security Classification Badge	Visual badge displaying account privilege (e.g. `Faculty (Active)`, `Student`, `Super Admin`).
NAME & EMAIL	User Identity Details	Displays verified full name and institutional email address used for login and notifications.
Show [10/25/50] Entries	Page Length Selector	Adjusts data table density for bulk audits.
Action Triggers (Edit / Reset)	Lifecycle Controls	Allows updating contact information, assigning new course roles, or issuing one-click password resets.

2.3 Bulk User Onboarding Center (CSV / Excel)

Allows rapid batch provisioning of hundreds of students, faculty, and administrators simultaneously with built-in schema validation and duplicate detection.

Figure 4: Bulk User Upload Drag & Drop Interface `Screenshot 2026-10-19 113136.png`



4-step automated pipeline (1. Upload > 2. Preview > 3. Confirm > 4. Result) with sample template download and duplicate detection.

4-Stage Onboarding Procedure

1. **Stage 1: Download Standard Template**

   Click **Download Template** to retrieve the pre-formatted CSV/Excel template containing mandatory headers: `Name`, `Email`, `Phone`, `Role`, and `Password`.

2. **Stage 2: Drag & Drop Ingestion**

   Drag your completed file into the designated upload zone. The system accepts `.csv`, `.xlsx`, and `.xls` files.

3. **Stage 3: Pre-Validation & Duplicate Check**

   Click **Preview & Validate**. The system scans for invalid email formats, missing mandatory fields, and existing database duplicates. Potential collisions are highlighted in amber.

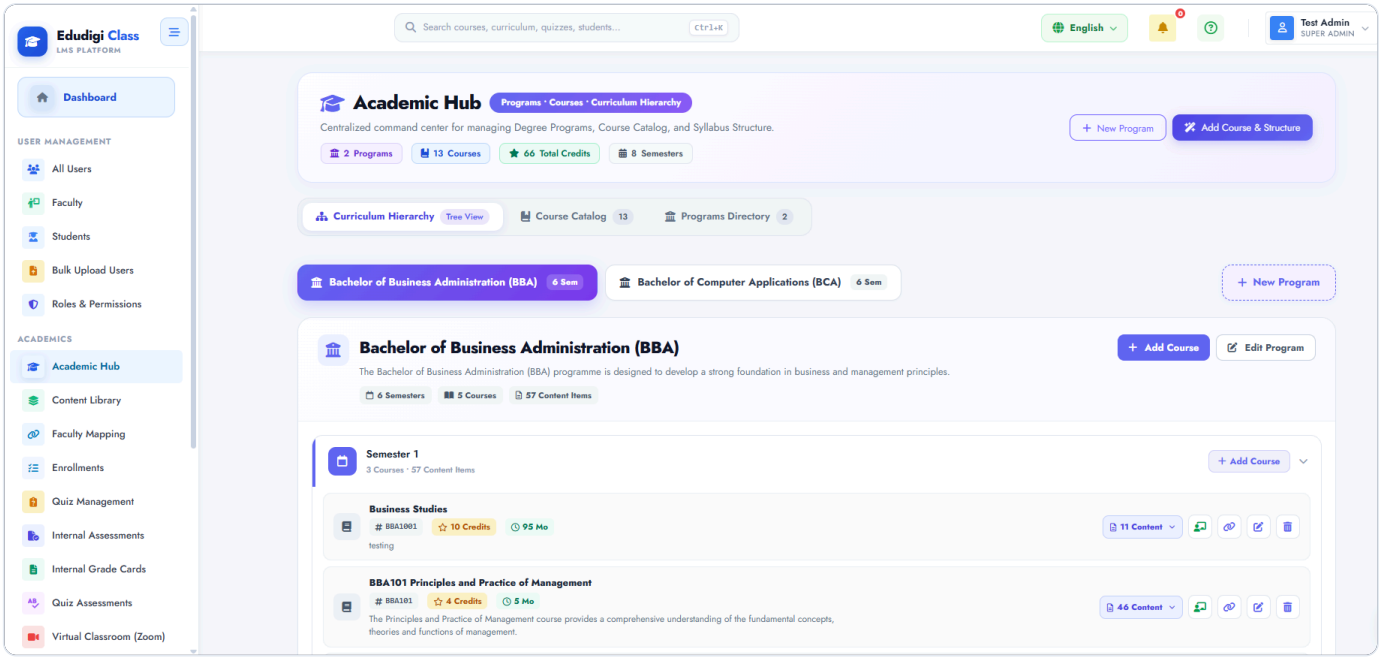
4. **Stage 4: Execution & Audit Receipt**

   Confirm creation. The system provisions the accounts, activates the **Force Password Change on First Login** flag, and generates a printable audit summary.

2.4 Academic Hub & Program Hierarchy

The central architectural command desk for establishing Degree Programs, Semester Structures, Subject Modules, and Credit Allocations.

Figure 5: Academic Hub & Curriculum Hierarchy `Screenshot 2026-10-19 113208.png`



Curriculum catalog tracking 2 Programs, 13 Courses, 66 Total Credits, and 8 Semesters with + New Program and + Add Course action triggers.

Bachelor of Business Administration (BBA)

6 Semesters • 5 Active Courses • 57 Content Items across business management, accounting, and marketing principles.

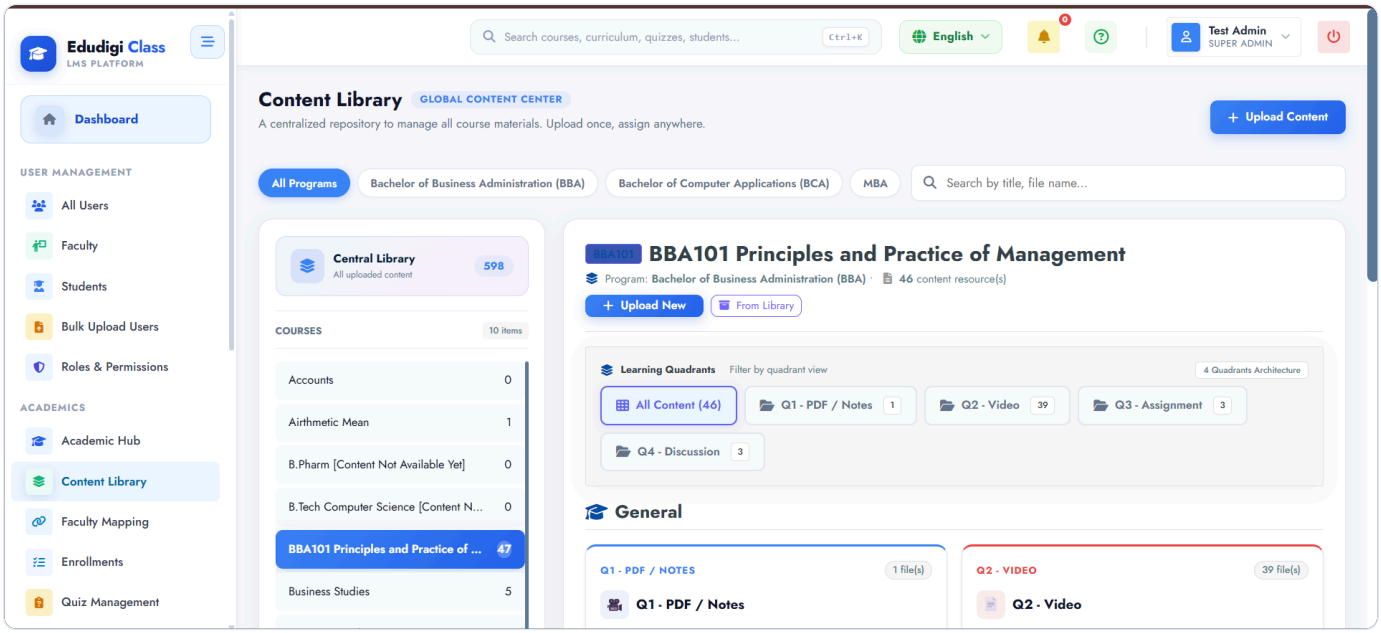
Bachelor of Computer Applications (BCA)

6 Semesters • 6 Active Courses • Computer networks, IoT, database management, and programming foundations.

2.5 Global Content Center (Content Library)

A unified digital repository to manage all multimedia learning assets. The architectural philosophy is **"Upload once, assign anywhere"**.

📸 **Figure 6: Global Content Center & Material Repository** `Screenshot 2026-10-19 113248.png`



Central repository enabling multi-program material uploads, quadrant tagging, and cross-course resource linking.

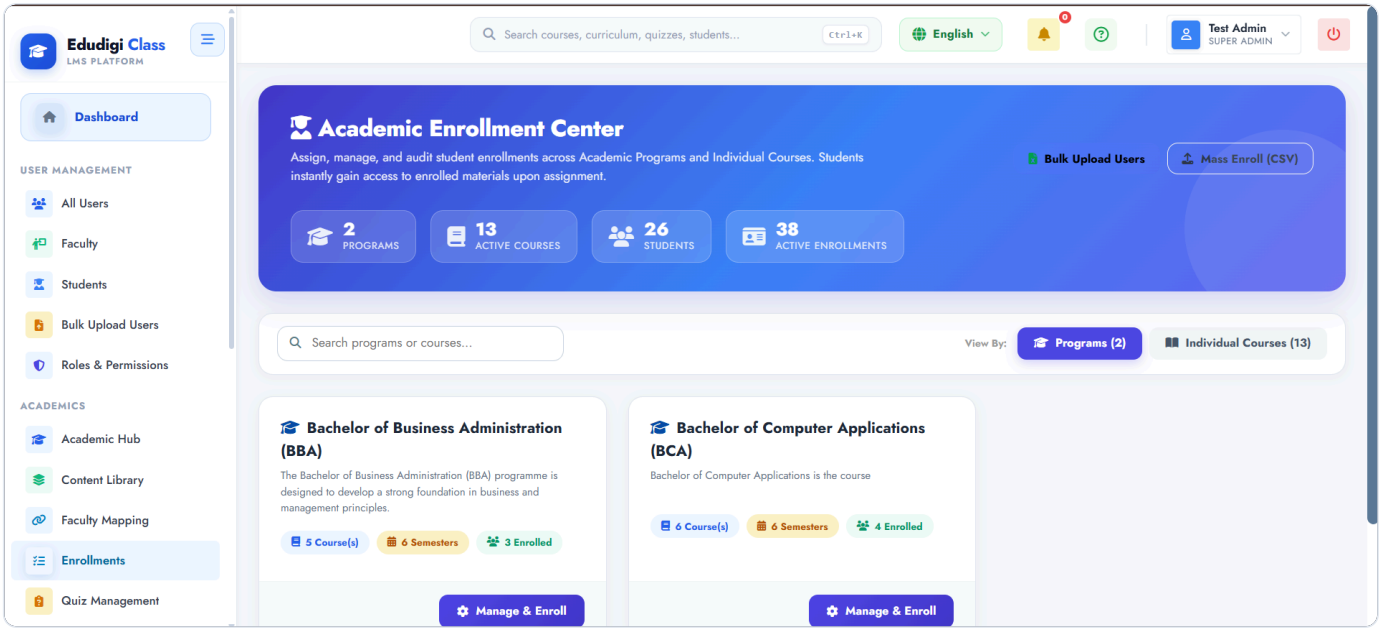
📦 **Storage & Verification Guard**

Uploaded PDFs, DOCX, and PPTX files undergo binary signature checks to prevent malicious file uploads. Files are indexed and rendered securely via the embedded document streaming viewer.

2.6 Academic Enrollment Center

Assign, manage, and audit student enrollments across Academic Programs and Individual Courses. Students instantly gain access to enrolled materials upon assignment.

Figure 7: Academic Enrollment Center `Screenshot 2026-10-19 113956.png`

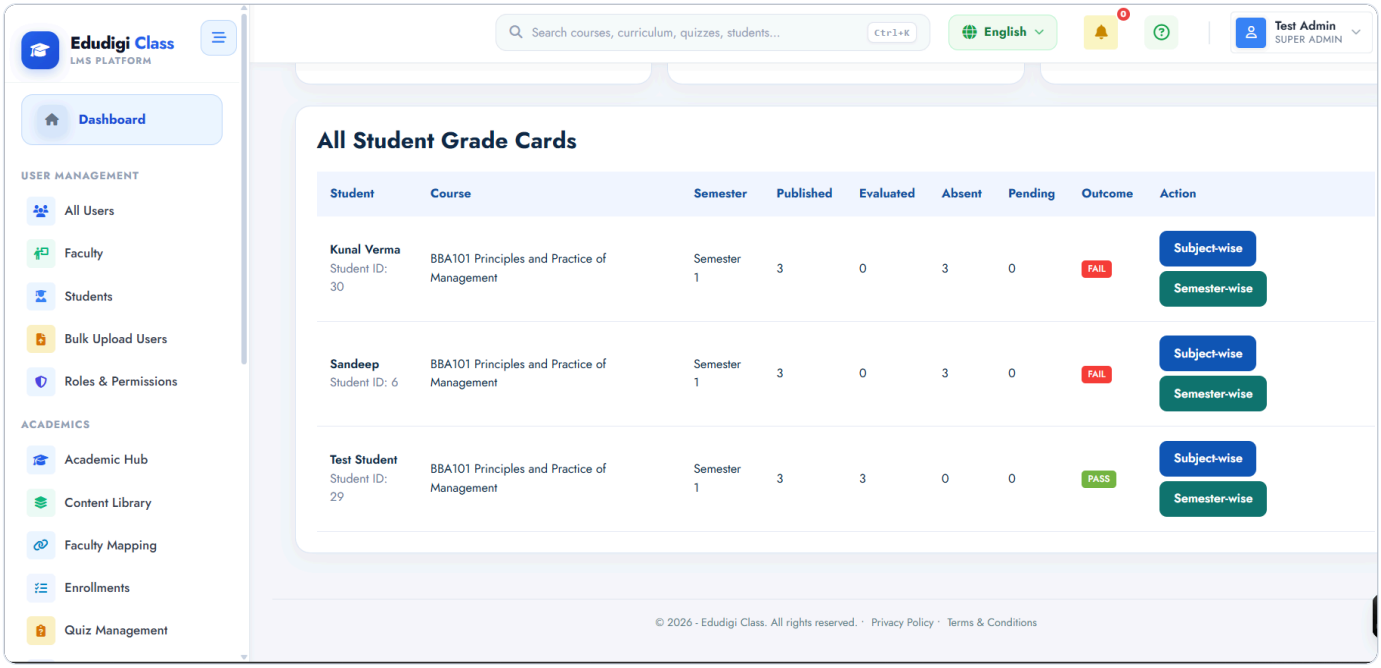


Enrollment cockpit monitoring 2 Programs, 13 Active Courses, 26 Students, and 38 Active Enrollments.

2.7 All Student Grade Cards & Evaluation Ledger

The institutional grade card repository tracks semester-wise student academic standings, marks earned across internal assessments, and UGC 10-point letter grades.

Figure 8: All Student Grade Cards Directory `Screenshot 2026-10-19 114250.png`

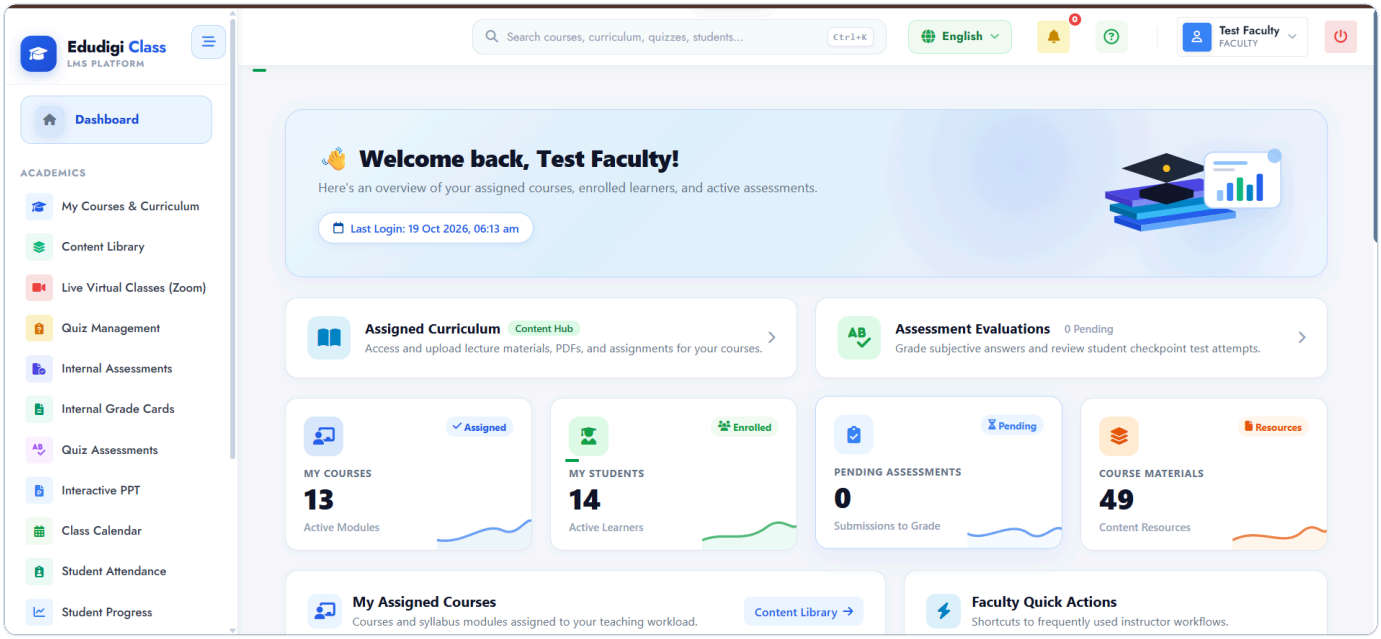


Grade card repository showcasing student profiles (e.g. Kunal Verma ID:30, Sandeep ID:6, Test Student ID:29), course titles, semesters, and assessment status badges (Published, Evaluated, Absent, Pending).

3.1 Faculty Teaching Command Desk

Instructors log in to manage their assigned curriculum, schedule live Zoom virtual lectures, author assessments, and evaluate student work.

Figure 9: Faculty Dashboard & Teaching Command Center `Screenshot 2026-10-19 114415.png`

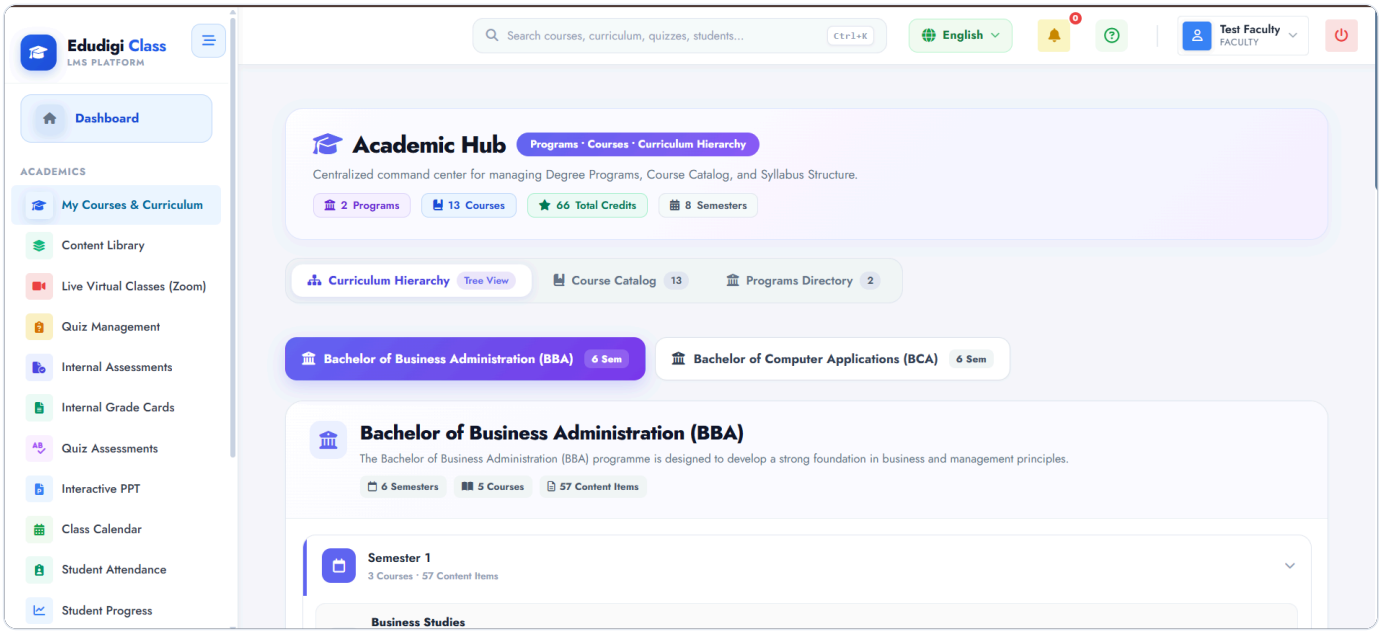


Faculty workspace displaying assigned courses, enrolled learners, upcoming virtual classes, and pending assessment evaluations.

3.2 Assigned Courses & Modular Curriculum

Faculty view the syllabus tree for their assigned subjects, organized into modular units, topics, and UGC 4-quadrant resources.

📸 **Figure 10: Faculty Assigned Curriculum & Course Hierarchy** `Screenshot 2026-10-19 114449.png`

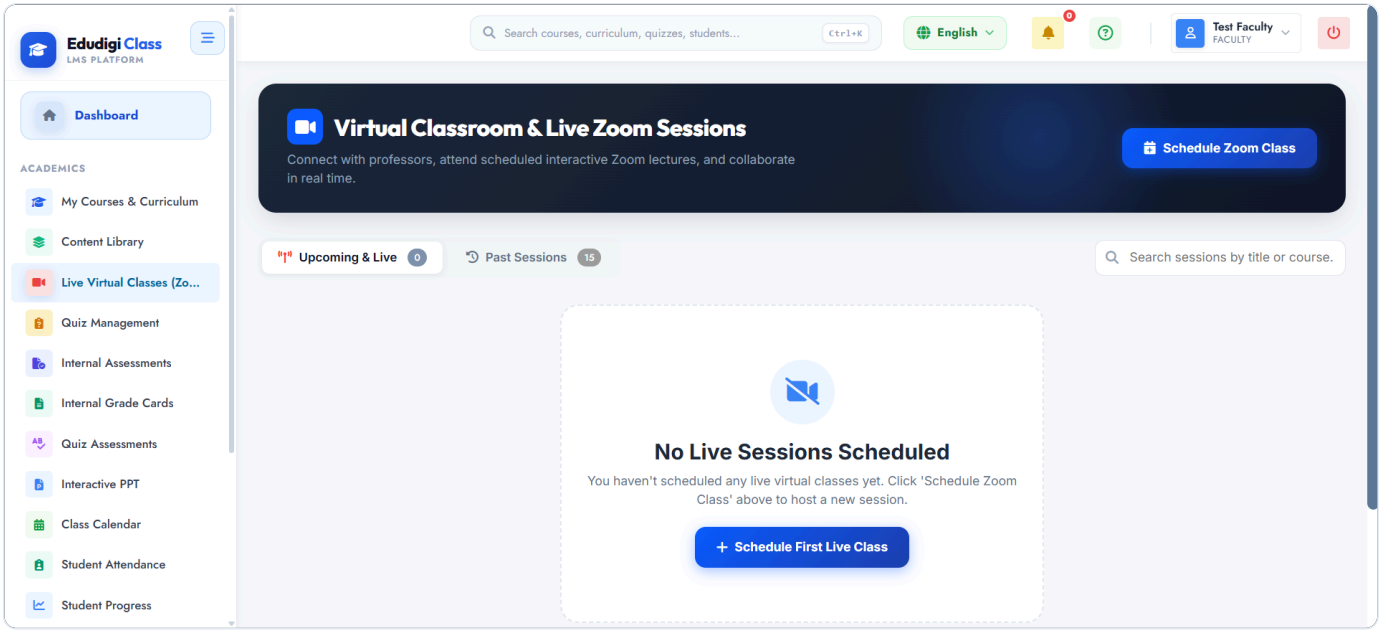


Curriculum catalog showing degree programs, course modules, credit allocations, and unit-level syllabus topics.

3.3 Virtual Classroom & Live Zoom Session Scheduling

Faculty schedule interactive lectures directly within LMS without needing to leave the platform. Secure meeting links are embedded directly into enrolled students' calendars.

Figure 11: Virtual Classroom & Live Zoom Sessions `Screenshot 2026-10-19 114514.png`



Live virtual lecture hub featuring the "Schedule Zoom Class" trigger, upcoming class roster, and automated attendance integration.

Step-by-Step Class Scheduling

1. **Click "Schedule Zoom Class"**
   Open the scheduling dialog on the Virtual Classroom page.
2. **Select Course & Topic**
   Choose your assigned course (e.g. *BBA101 Principles and Practice of Management*), specify the topic title, and select date and time.
3. **Automated Meeting Creation & Roster Broadcast**
   The system provisions the meeting and immediately updates the Class Calendar of every enrolled student with a 1-click **Join Class** button.

3.4 AI Question Bank & Automatic Question Paper Generator

The platform features an advanced **Google Gemini AI** question generation engine and a formal **Question Paper Generator** adhering to institutional exam blueprints.

Curriculum-Grounded AI Questions

Generate curriculum-aligned MCQs, Short Answer, and Long Descriptive questions filtered by Bloom's Taxonomy (Remember, Understand, Apply, Analyze, Evaluate, Create) and difficulty (Easy, Medium, Hard).

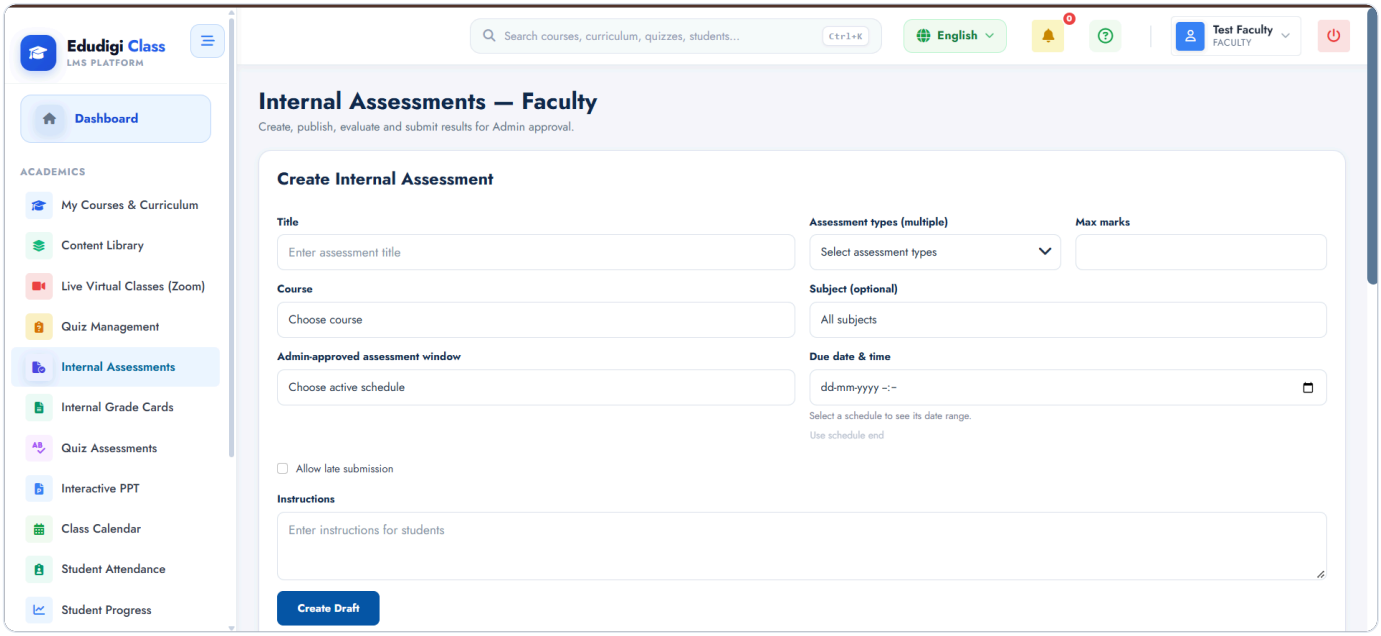
1-Click Blueprint Paper Auto-Assembly

Auto-assemble structured exam papers with configurable sections (Section A: MCQs, Section B: Short Notes, Section C: Long Problems) and print institutional hardcopies.

3.5 Internal Assessment Policy & Window Configuration

Faculty create formal assessments aligned with administrator-approved assessment windows, configuring submission types, deadlines, and passing thresholds.

Figure 12: Internal Assessment Creation Modal `Screenshot 2026-10-19 114550.png`



Authoring form with Title, Max Marks, Course selector, Admin-approved window, Multiple Assessment Types, and Late Submission toggle.

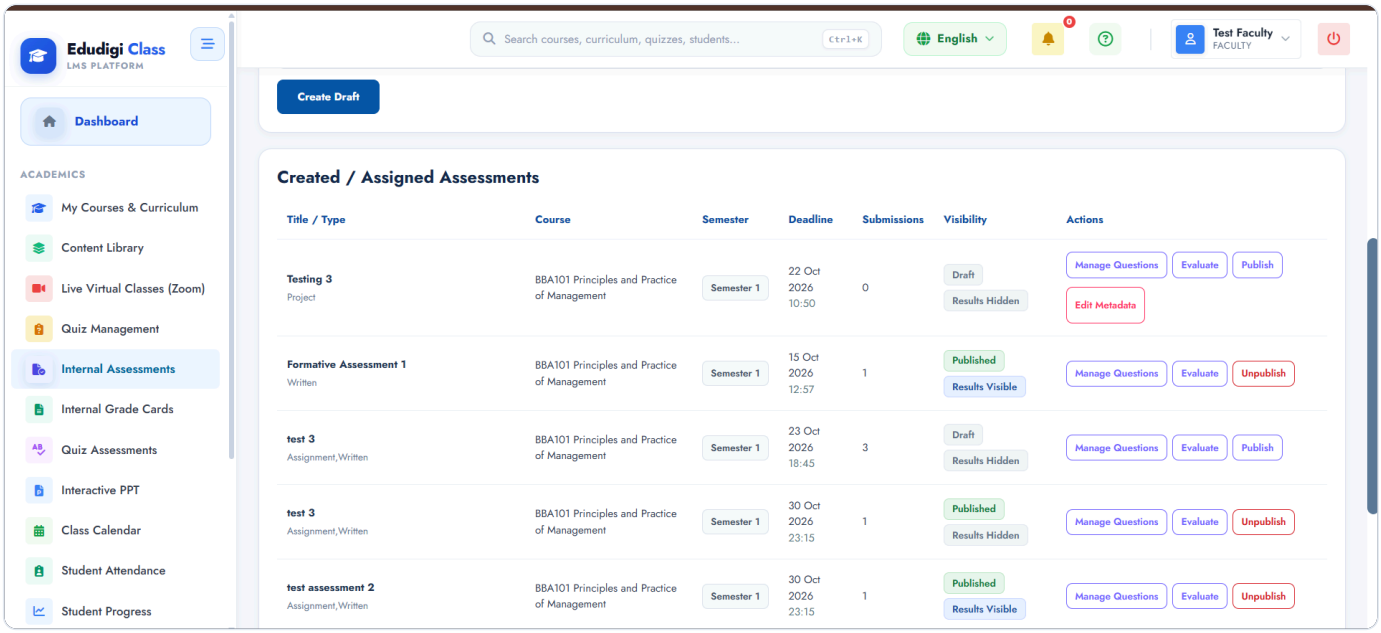
Assessment Field Specification

FIELD / OPTION	TYPE	DESCRIPTION & BUSINESS RULES
Title	Text Input	Descriptive name of the assessment (e.g. *Formative Assessment 1 - Unit 2 Written Test*).
Course	Dropdown	Selects the target course. Only courses mapped to the logged-in faculty appear in this list.
Admin Approved Window	Schedule Selector	Links the assessment to an authorized university window ensuring compliance with term dates.
Assessment Types (Multiple)	Multi-Select Picker	Select one or more: `Assignment`, `Written`, `Practical`, `Project`, `Presentation`, `Viva`, `MCQ`.
Max Marks	Numeric Input	Total maximum marks (e.g. 100.00). Total question marks must equal this value upon publication.
Allow Late Submission	Checkbox	If enabled, submissions after the due date are flagged as "Late" but accepted for grading.

3.6 Assessment Drafting & Question Authoring

Manage drafts, configure rubrics, and define expected answers for Gemini AI evaluation.

📸 **Figure 13: Created & Assigned Assessments Roster** `Screenshot 2026-10-19 114657.png`



Assessment management table displaying Title, Type (Project, Written, Assignment), Course, Semester, and Action buttons (Submissions, Grade, Questions).

Gemini AI Descriptive Grading Engine

EduDigiClass integrates a state-of-the-art Google Gemini AI evaluation engine:

50% Conceptual Benchmark Threshold

Evaluates understanding depth. Submissions matching ≥ 50% of core concepts are flagged as Benchmark Met. Below 50% triggers mandatory faculty review.

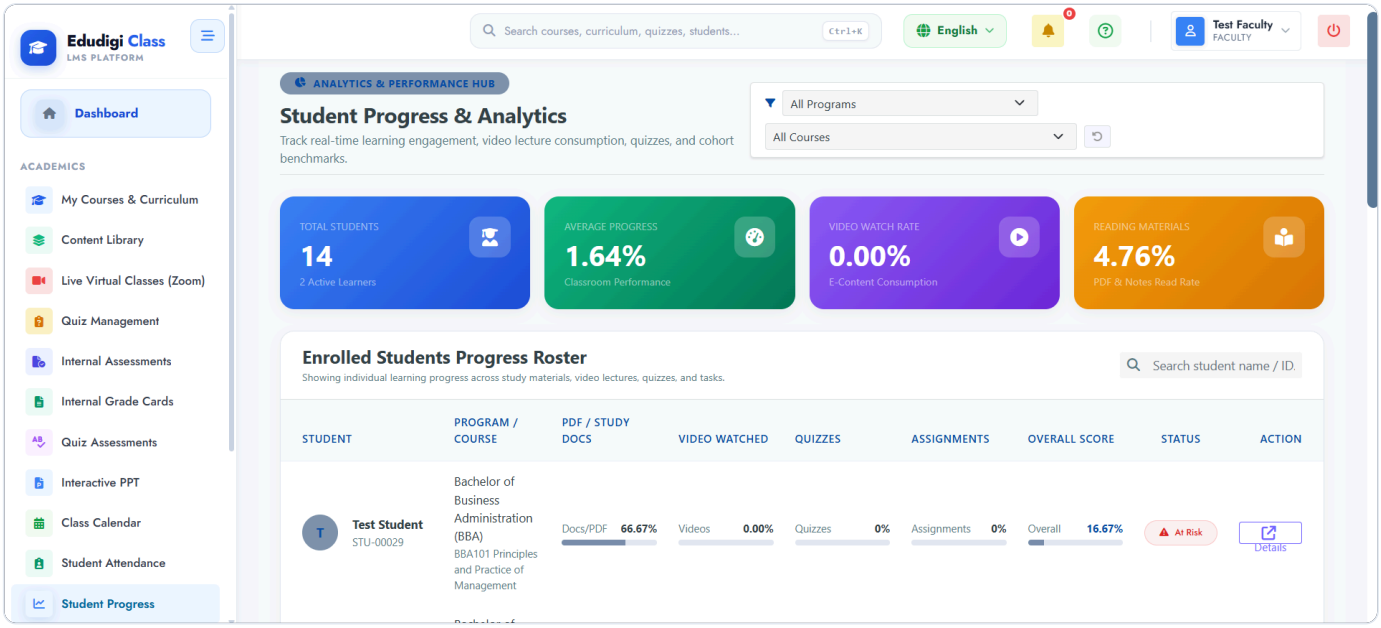
Natural Academic Marks Ceiling

Prevents unrealistic 100% scores on subjective essays. Implements realistic increments (.00, .25, .50, .75) with natural ceilings (80–88% for exceptional work).

3.7 Learner Telemetry & Performance Analytics Hub

Instructors track cohort engagement in real time, inspecting video lecture watch time, reading completion, and identifying at-risk learners.

Figure 14: Student Progress & Learning Analytics Hub `Screenshot 2026-10-19 114732.png`

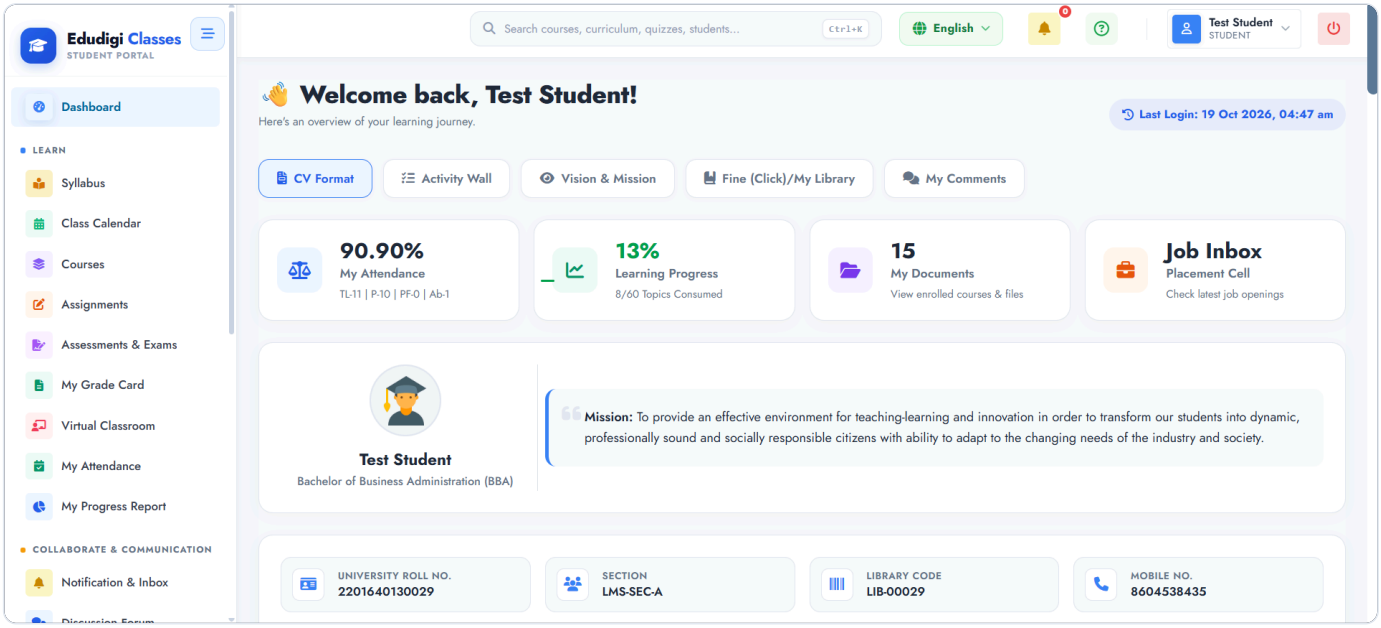


Analytics cockpit displaying program-wide engagement metrics, video lecture consumption percentages, quiz scores, and at-risk learner highlights.

4.1 Student Portal Dashboard & Academic Journey

The student dashboard is the personalized home for every learner, providing instant visibility into daily schedules, course progress, and announcements.

Figure 15: Student Portal Dashboard Overview `Screenshot 2026-10-19 114812.png`



Student homepage showing "Welcome back, Test Student!", Last Login timestamp (19 Oct 2026, 04:47 am), enrolled courses, and navigation drawer.

Student Navigation Structure

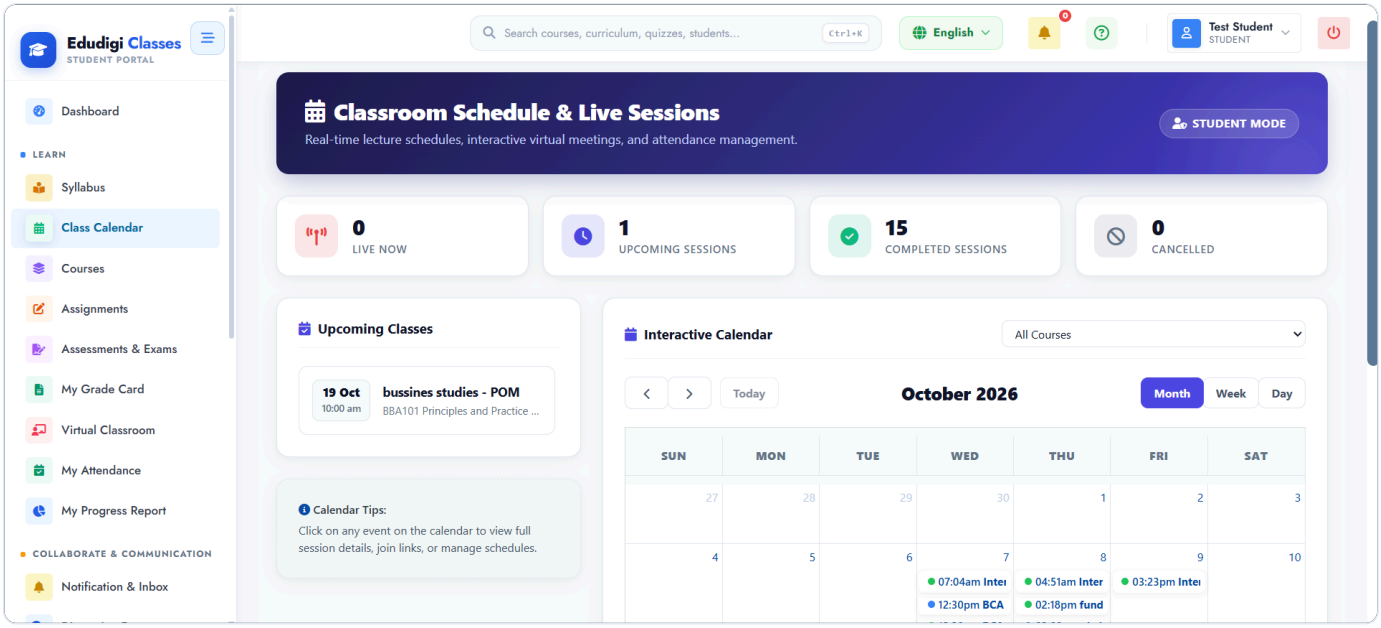
MENU ITEM	SECTION	FUNCTION & TARGET DESTINATION
Dashboard	LEARN	Home view with summary cards, course progress widgets, and quick-start links.
Syllabus	LEARN	Official digital syllabus outlines, course codes, and credit breakdowns.
Class Calendar	LEARN	Schedule of all upcoming live Zoom classes, assignment deadlines, and exam dates.
Courses	LEARN	UGC 4-Quadrant multimedia learning hub for all actively enrolled subjects.
Assignments & Exams	LEARN	Assessment taking center for formative tests, quizzes, and end-term examinations.
My Grade Card	LEARN	Interactive dashboard and A4 printable semester grade card with UGC 10-point scale.
My Attendance	LEARN	Detailed login presence ledger tracking attendance percentage and daily logins.
My Progress Report	LEARN	Telemetry breakdown across video watch time, reading completion, and quiz scores.

STUDENT MANUAL

4.2 Classroom Schedule & Live Zoom Sessions

Students access daily virtual classes via the interactive schedule. With 1-click Zoom integration, learners join live lectures directly from their browser or mobile app.

Figure 16: Student Classroom Schedule & Live Sessions `Screenshot 2026-10-19 114835.png`

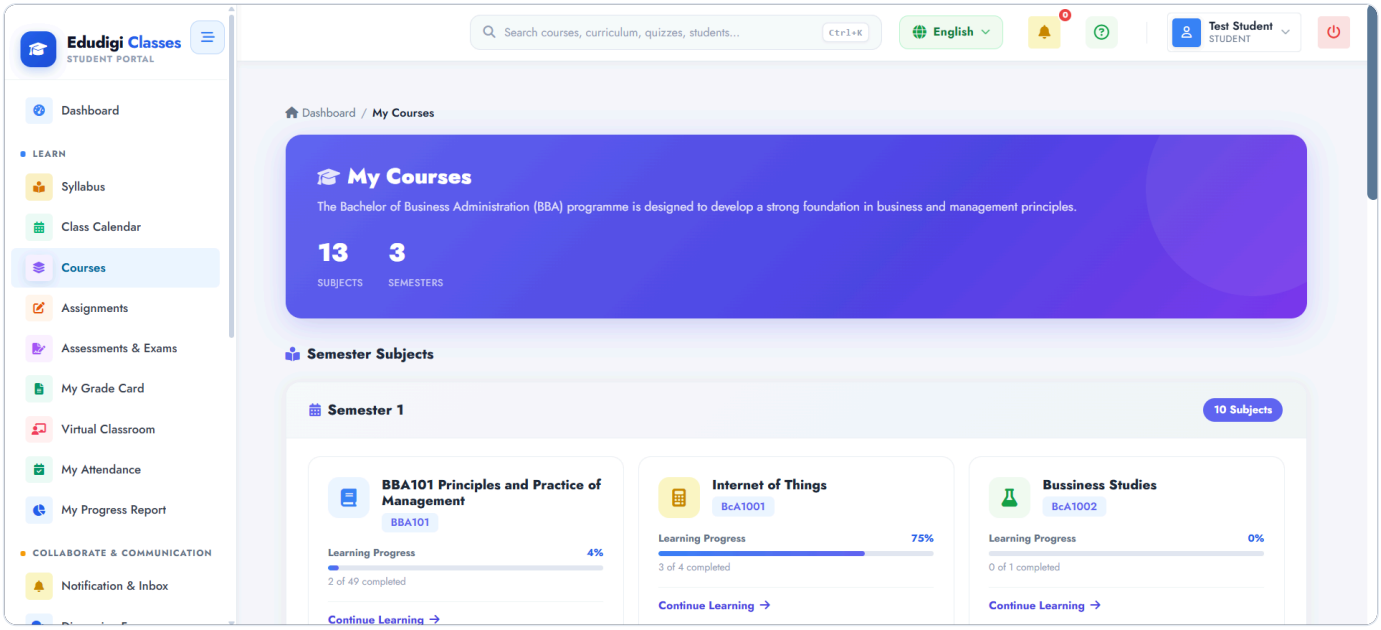


Classroom calendar showing scheduled live meetings, start times, course names, and the direct Join Class button.

4.3 My Courses & UGC 4-Quadrant Learning Hub

All subjects follow the statutory **UGC 4-Quadrant E-Learning Framework**, organizing curriculum content into four distinct modes of instruction.

Figure 17: My Courses & Quadrant Learning Hub `Screenshot 2026-10-19 114904.png`



Enrolled course viewer displaying subject cards, progress indicators, syllabus units, and Continue Learning action buttons.

Quadrant 1: E-Tutorial (Video)

High-definition video lectures and recorded sessions with automatic watch time tracking and bookmarking.

Quadrant 2: E-Content (Reading)

Digital textbooks, downloadable PDF notes, case studies, and research compendiums rendered securely.

Quadrant 3: Self-Assessment

Topic quizzes, multiple-choice practice tests, and self-check questions with instant answer explanations.

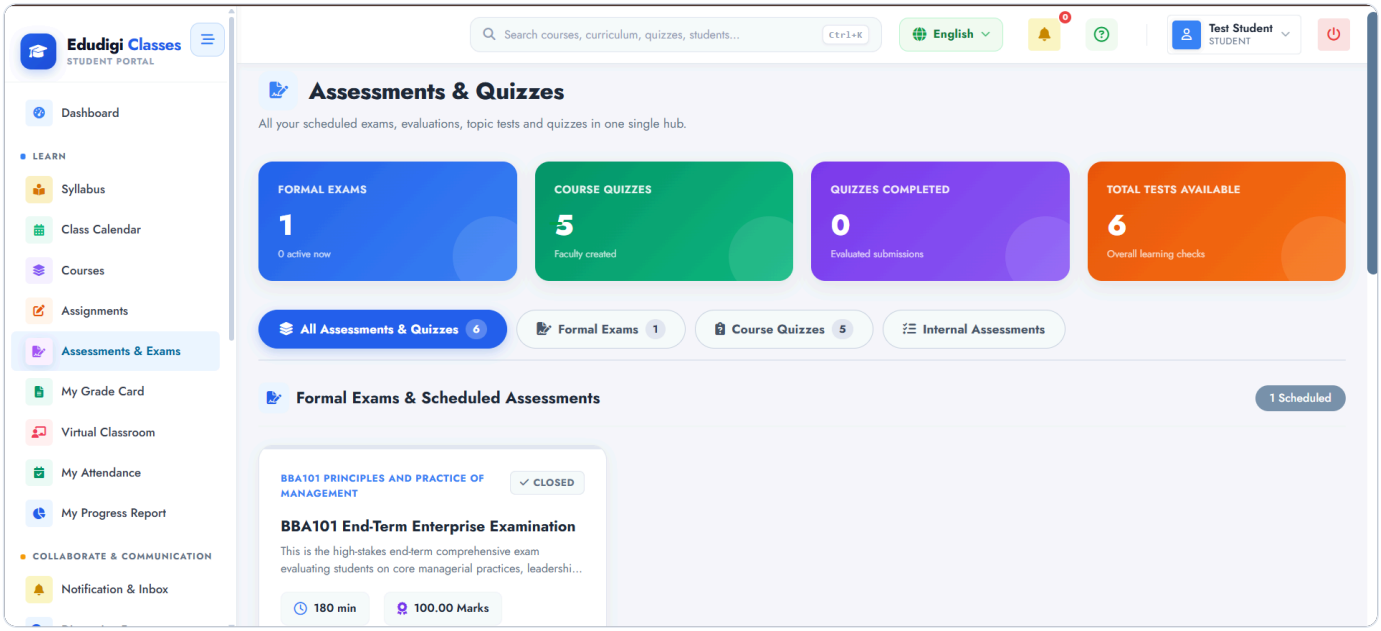
Quadrant 4: Discussion Forum

Moderated course discussion boards, doubt-clearing threads, and peer collaboration with instructors.

4.4 Assessments, Quizzes & Exam Taking Center

Learners attempt scheduled topic tests, submit written project files, and sit for formal semester examinations within a secure timed browser environment.

Figure 18: Assessments & Quizzes Center `Screenshot 2026-10-19 114935.png`



Unified assessment desk showing scheduled tests, duration, total marks, deadlines, and the "Attempt Assessment" trigger.

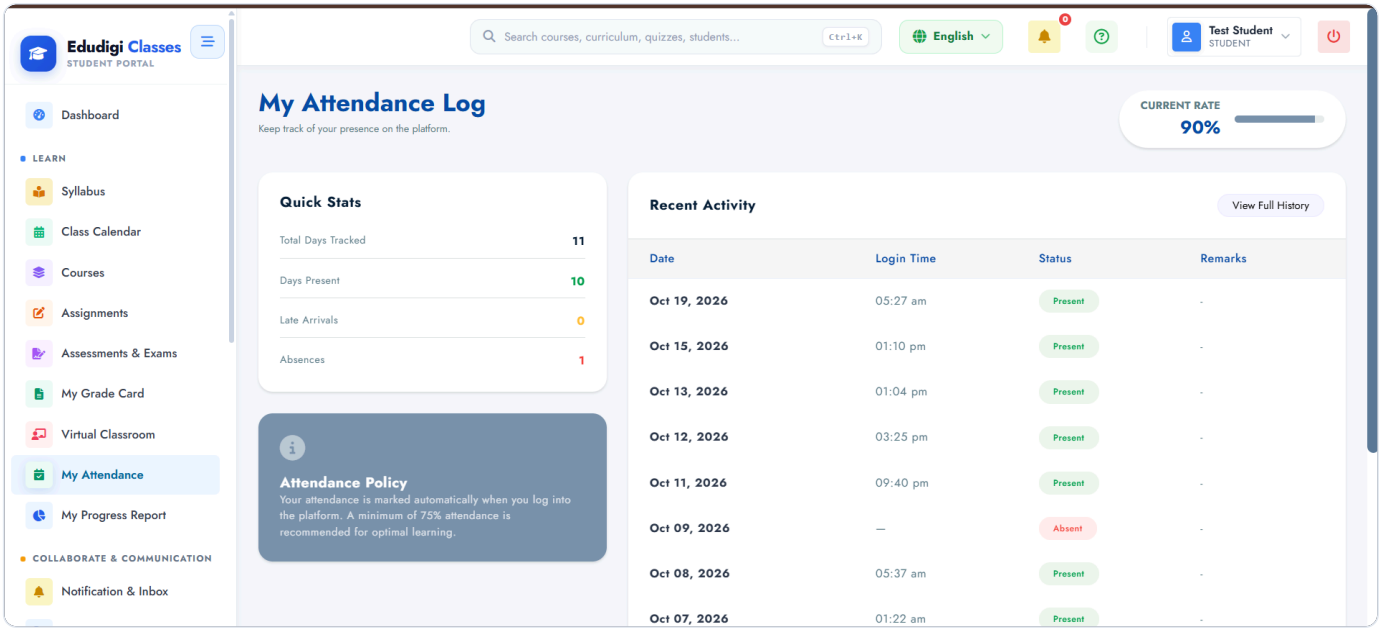
Timed Assessment Security Rules

Once initiated, the countdown timer runs continuously. Auto-save ensures answers are preserved even if the connection drops. Students submitting Project or Presentation tasks can upload PDF/DOCX files up to 8MB.

4.5 Login-Based Automatic Attendance Telemetry

In addition to lecture presence, EduDigiClass features automatic login attendance logging. Whenever a student logs into the platform, their presence is recorded in the immutable institutional ledger.

Figure 19: My Attendance Log & Telemetry Ledger `Screenshot 2026-10-19 115010.png`



Attendance telemetry displaying current attendance rate (90.9%), Days Tracked (11), Days Present (10), Absences (1), and daily login timestamps.

Attendance Policy Rules

TELEMETRY METRIC	CURRENT VALUE	COMPLIANCE STANDARD
Current Attendance Rate	**90.9%**	Minimum 75% required by statutory university policy to qualify for end-term examinations.
Total Days Tracked	11 Days	Total instructional days elapsed since term enrollment.
Days Present	10 Days	Days where successful login session was verified.
Late Arrivals / Absences	0 Late / 1 Absent	Unrecorded or missed instructional days.

4.6 Progress Report & Grade Card Download

Students monitor overall learning metrics across videos, reading materials, quizzes, and assignments, with instant A4 PDF grade card generation.

Figure 20: Test Student's Progress Report

`Screenshot 2026-10-19 115033.png`



Performance breakdown across all active subjects showing quiz scores, assignment grades, and video watch telemetry.

5.1 Institutional UGC 10-Point Letter Grading Scale

The platform implements the statutory 10-point Choice Based Credit System (CBCS) grading scale:

PERCENTAGE BAND	LETTER GRADE	GRADE POINT	PERFORMANCE CLASSIFICATION	ACADEMIC STANDING
≥ 90.00%	O	10.0	Outstanding	Cleared with Distinction
80.00% – 89.99%	A+	9.0	Excellent	Cleared
70.00% – 79.99%	A	8.0	Very Good	Cleared
60.00% – 69.99%	B+	7.0	Good	Cleared
50.00% – 59.99%	B	6.0	Above Average	Cleared
40.00% – 49.99%	C	5.0	Average	Cleared
33.00% – 39.99%	P	4.0	Pass	Marginal Pass
< 33.00%	F	0.0	Fail	Eligible for Compartmental Re-assessment

⚖️ **Grace Marks & Compartmental Examination Engine**

Students failing within a narrow threshold (1–5 marks) can receive automated discretionary grace marks as governed by institutional policy. For students receiving an **F** grade, the system enables 1-click creation of a compartmental supplementary assessment (Attempt #2).

REFERENCE & SUPPORT

5.2 Frequently Asked Questions & Troubleshooting

Q: My Zoom live class link is not opening or shows "Host has not started meeting".

A: Live classes require the faculty host to start the room first. If you join earlier than 5 minutes before scheduled start time, wait on the meeting room page or refresh your browser.

Q: My file upload for the project assignment was rejected.

A: Ensure your file is in `.pdf`, `.docx`, `.pptx`, or image formats, and does not exceed the **8 MB** limit. Encrypted or password-protected files cannot be processed by the server validator.

Q: How do I print my official semester grade card?

A: Navigate to **My Grade Card** in the student menu. Click the blue **Print Grade Card** button. In the browser print dialog, select **Save as PDF** with paper size set to **A4 Portrait**.

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